

SUCHITH CHIDANANDA PRABHU

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EDUCATION

Indian Institute of Technology Delhi

Ph.D. in Artificial Intelligence · GPA: 9.4

Delhi, India
Aug 2021 – Present

- **Research:** Memory-Augmented Models for Efficient Extreme Classification and Dense Retrieval
- **Coursework:** Information Retrieval (A) · NLP (A–) · Data Mining (A) · Numerical Algorithms (A) · Machine Learning (B)

National Institute of Technology Goa

Bachelor of Computer Science · GPA: 9.25 (First in class of 30)

Goa, India
Aug 2015 – May 2019

PUBLICATIONS

ElasticSkip: Where to Cut? Contiguous Middle-Skip Beats Importance Pruning for Dense-Retrieval Query Encoders

Submitted to KDD 2027

Dense retrieval, Model compression

- Compresses LLM-based query encoders by skipping a contiguous block of middle layers, preserving the final layers that refine the pooled query embedding toward concept-level meaning.
- Depth-specific low-rank adapters and a denoised-contrastive recipe let a single base model cover the full depth–quality trade-off, with no retraining or re-indexing per latency target.
- On MS MARCO, improves nDCG@10 by up to **1.7** pts over norm-ratio and **4.1** pts over Block-Influence pruning; matches or beats SOTA HARNESS-LM at every depth at 0.6B, 4B and 8B scale.

MOGIC: Metadata-Infused Oracle Guidance for Improved Extreme Classification

ICML 2025

Extreme classification · First author · github.com/suchith720/mogic

- Two-phase training: an early-fusion oracle with access to query- and label-side ground-truth text metadata guides memory-based XC models via regularization, combining early- and late-fusion.
- Improves P@1 and PSP@1 by **1–2%** on six benchmarks (up to 1.3M labels) at **no additional inference cost**; robust to missing and noisy metadata.
- Plug-and-play: also boosts memory-free XC models such as NGAME and DEXA.

RAMEN: Graph Regularized Encoder Training for Extreme Classification

WWW 2025

Extreme classification, Sponsored search

- Uses graph metadata (e.g., co-session queries) as a regularizer for a lightweight encoder instead of a compute- and memory-heavy GCN, avoiding noisy graph induction on tail queries and ads.
- A/B tests on live multi-lingual Bing Ads traffic: **+1.25–1.5%** revenue and **+0.5–0.6%** CTR across markets.
- On public benchmarks, up to 5% higher accuracy than SOTA with 50% faster inference and 70% fewer parameters.

EXPERIENCE

Microsoft Research India

Business Guest

Bengaluru, India
Jan 2023 – Present

- Led research on **MOGIC** (ICML 2025) and **ElasticSkip** (submitted to KDD 2027); collaborated on **RAMEN** (WWW 2025).

Vehant Technologies

Software Developer – Under-Vehicle Scanning System

Noida, India
Jul 2019 – Jul 2021

- Led stereo fisheye camera calibration and rectification (Blackfly S GigE) for the scanning rig.
- Implemented ELAS stereo matching for dense depth estimation of vehicle undersides.
- Built automated foreign-object detection via image comparison, and a stitching pipeline for full under-vehicle images.

Indian Academy of Sciences Summer Research Fellowship

Research Intern – Vijnana Labs

Bengaluru, India
May 2018 – Jul 2018

- Developed an AI framework to detect adulterants in pharmaceuticals and oils using infrared spectrum signature analysis.

R. C. Bose Cryptology Internship

Research Intern – Indian Statistical Institute

Kolkata, India
May 2017 – Jul 2017

- Implemented and analyzed consensus algorithms (Byzantine, Phase-King, Paxos) for decentralized systems.
- Built an Ethereum dApp combining cloud storage and blockchain data structures for secure, auditable storage.

HONORS & AWARDS

Prime Minister's Research Fellowship (PMRF)

Awarded by the Government of India for excellence in research and academic achievement

Jan 2023 – Present